

ANUCHINO, Russian Federation

WMO: 319810

Lat: 43.967N

Lon: 133.067E

Elev: 615

StdP: 14.37

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.3	-19.6	-28.0	1.2	-22.2	-24.4	1.5	-18.5	12.3	5.8	10.0	4.7	5.0	250	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	87.6	71.4	84.5	69.7	81.8	68.8	74.3	83.4	72.7	80.9	71.1	78.2	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	118.6	79.2	70.0	113.1	77.5	68.6	107.6	75.6	38.4	83.7	37.0	80.7	35.5	78.3	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	9.7	8.7	-27.7	92.4	4.9	2.9	-31.3	94.5	-34.1	96.2	-36.9	97.8	-40.4	100.0		
			-27.9	77.4	4.8	2.4	-31.4	79.1	-34.3	80.5	-37.0	81.9	-40.5	83.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	0.0	9.1	25.3	43.0	55.8	64.8	71.3	70.1	59.0	44.3	24.1	5.2
	DBStd	26.11	8.74	10.13	10.34	7.85	6.51	5.52	4.70	4.88	6.35	8.02	11.96	10.06
	HDD50	6101	1549	1145	764	238	20	0	0	0	8	211	778	1388
	HDD65	9770	2014	1565	1229	662	292	71	6	12	197	642	1227	1853
	CDD50	2266	0	0	0	27	200	444	662	622	277	34	0	0
	CDD65	459	0	0	0	0	7	64	203	169	16	0	0	0
	CDH74	3634	0	0	0	27	219	686	1488	1077	133	4	0	0
	CDH80	994	0	0	0	4	49	187	478	262	14	0	0	0
	WSAvg	4.0	4.0	4.3	4.5	4.9	4.3	3.6	3.1	3.1	3.5	4.1	4.0	3.9
	PrecAvg	28.0	0.5	0.6	1.0	1.7	2.9	3.1	4.8	5.0	3.6	2.6	1.4	0.9
(15) Precipitation	PrecMax	40.3	1.2	1.7	2.2	4.0	5.4	6.6	9.2	12.8	10.2	6.2	2.7	2.0
	PrecMin	16.2	0.0	0.0	0.0	0.3	0.2	1.2	1.5	0.7	0.8	0.6	0.4	0.1
	PrecStd	5.0	0.3	0.5	0.6	0.9	1.2	1.4	1.8	2.6	2.6	1.3	0.6	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	43.3	58.2	78.1	85.8	89.1	92.6	89.9	81.8	73.0	58.6	41.2
		MCWB	29.8	35.2	43.6	54.6	61.3	68.6	74.7	74.2	67.0	57.6	46.8	36.8
	2%	DB	25.6	37.6	50.3	70.2	79.8	85.0	88.4	85.8	77.7	67.3	52.0	32.6
		MCWB	22.5	31.5	38.5	50.3	58.7	67.1	72.6	72.7	64.6	53.6	43.2	29.1
	5%	DB	19.3	32.4	45.2	64.1	75.4	81.5	85.1	82.8	74.5	62.6	46.1	26.4
		MCWB	16.6	28.0	35.8	47.6	56.6	65.9	71.2	71.3	62.7	51.4	39.9	23.7
	10%	DB	15.0	27.2	40.6	59.0	70.9	77.7	82.1	80.1	71.2	58.6	40.5	21.2
		MCWB	12.8	23.2	33.5	45.2	54.7	64.2	70.6	70.0	60.9	49.9	35.4	19.0
	0.4%	WB	29.8	36.3	44.5	56.3	63.4	71.7	76.9	76.3	71.1	59.3	48.1	36.9
		MCDB	33.1	41.5	57.1	73.9	78.6	82.9	88.8	86.2	78.3	68.8	56.6	41.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	22.6	32.2	39.9	51.7	60.3	69.4	74.6	74.5	67.3	56.1	44.4	29.2
		MCDB	25.7	36.8	48.3	67.4	75.5	80.5	84.1	83.2	73.2	63.8	50.8	32.2
	5%	WB	16.9	27.7	36.8	48.6	58.2	67.5	73.1	73.0	64.8	53.7	40.0	23.8
		MCDB	19.3	32.4	43.9	62.1	72.3	78.2	82.0	80.5	71.4	60.2	45.7	26.4
	10%	WB	12.9	23.1	34.1	45.7	56.2	65.4	71.5	71.3	62.8	50.8	35.8	18.9
		MCDB	14.9	27.2	39.9	57.1	68.1	75.5	79.3	77.4	69.2	57.6	40.2	21.2
(35) Mean Daily Temperature Range	5% DB	MDBR	25.5	27.0	22.0	22.7	23.1	21.0	17.4	17.0	22.3	23.4	21.2	23.1
		MCDBR	28.4	29.5	26.6	33.4	32.8	28.4	24.0	21.4	26.0	30.7	27.0	25.2
		MCWBR	25.0	23.2	16.4	17.2	14.2	12.8	9.9	9.4	13.0	17.3	18.2	21.7
	5% WB	MCDBR	28.0	28.6	24.2	30.9	28.2	24.7	19.8	17.9	20.7	26.5	26.1	24.9
		MCWBR	24.8	23.0	15.6	16.8	13.1	12.0	9.0	8.7	11.3	16.3	18.3	21.7
(40) Clear-Sky Solar Irradiance	taub		0.325	0.387	0.472	0.572	0.580	0.552	0.516	0.475	0.429	0.445	0.390	0.344
	taud		2.066	1.893	1.788	1.674	1.718	1.870	2.033	2.148	2.186	1.997	2.035	2.028
	Ebn at Noon		241	241	236	223	227	235	242	248	250	220	212	216
	Edh at Noon		33	48	61	74	73	62	53	45	40	43	35	31
(44) All-Sky Solar Radiation	RadAvg		680	995	1316	1454	1618	1717	1536	1451	1311	952	655	549
	RadStd		42	53	91	140	132	188	173	134	110	82	58	33

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page